

How to start/adjust RR-TB treatment

ABC – abacavir; **AZT** – zidovudine; **BDQ** – bedaquiline; **CFZ** – clofazimine; **DTG** – dolutegravir; **EFV** – efavirenz; **FLQ** – fluoroquinolone; **FTC** – emtricitabine; **LPVr** – lopinavir/ritonavir; **LZD** – linezolid; **TDF** – tenofovir; **3TC** – lamivudine

STEP 1: If any of the following, refer to hospital for admission

- Respiratory rate > 20
- Hb < 8
- BMI < 18
- Suspected TB meningitis or brain tuberculoma
- Unable to walk unaided
- Unstable social circumstances
- Difficulty with adherence
- Patient requests admission
- Infection control challenges at home

STEP 2: If starting treatment as outpatient or hospital admission not possible, decide which RR-TB regimen to give

Does patient have any of:

- Hb < 8
- Complicated EPTB¹
- Extensive bilateral cavitations on chest x-ray
- Previous RR-TB treatment for > 1 month
- Both inhA and KatG mutations on LPA
- Patient is a TB contact² of index patient with both inhA and katG mutations
- Patient is a TB contact² of index patient with resistance to FLQ, injectables, BDQ, LZD or CFZ
- Patient is a TB contact² of index patient failing treatment

None of above

Start **shorter** regimen ↪ 104.

One or more of above

Does patient have any of:

- Hb < 8
- QTcF ≥ 450ms
- Patient is a TB contact² of index patient failing treatment
- CNS disease (TB meningitis or brain tuberculoma)
- Patient is a TB contact² of index patient with resistance to FLQ, BDQ, LZD or CFZ

No

Start **basic longer** regimen ↪ 104.

Yes

- Discuss **individualised longer** regimen with PCAC/NCAC.
- Follow up LPA and phenotypic DST results and discuss.

Review LPA and phenotypic DST results:

- If discordance³ or heteroresistance⁴: continue same regimen and discuss with laboratory and PCAC/NCAC.
- Does patient have both inhA and katG mutations on LPA, or resistance to FLQ, BDQ, LZD or CFZ?

No

- Continue same regimen.
- If on shorter regimen and INH susceptible on both LPA and phenotypic DST, reduce high dose INH to normal dose INH ↪ 104.

Yes

- If resistance to FLQ, BDQ, LZD or CFZ: discuss **individualised longer** regimen with PCAC/NCAC.
- Otherwise continue/change to **basic longer** regimen ↪ 104.

STEP 3: If on ART, adjust ART regimen

- Avoid giving EFV and BDQ or, AZT and LZD, together because of overlapping toxicities.
- Check if patient eligible to switch same day to TDF + 3TC + DTG (TLD) ↪ 117.

¹TB meningitis or brain tuberculoma/TB spine/bone/joint or miliary, pericardial, abdominal or urogenital TB. ²A TB contact refers to a patient who shared an enclosed space (at work, socially, in a hostel, or in a household setting), for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index patient"), during the 3-month period before the index patient started their TB treatment. ³Discordance here refers to instance where TB NAAT result is rifampicin-resistant and DST result is rifampicin-sensitive. ⁴Heteroresistance here refers to both rifampicin-susceptible and rifampicin-resistant strains of TB in the same sputum sample.